

Lab: Thought Experiment — Planning, Time, and Purpose

Objective

Investigate the philosophical foundations of planning through thought experiments on temporal depth, purpose, and the relationship between planning and human flourishing.

Part 1: The Chess Player's Mind

Goal: Analyze planning depth and counterfactual reasoning.

A chess grandmaster evaluates a position. She considers: "If I move the knight here, my opponent will likely respond with X, which opens up Y, which after three more moves gives me a strategic advantage."

- How many levels of "sophisticated inference" are involved? Map out the recursive structure.
- How does this compare to the planning of a novice who can only evaluate one move ahead?
- Does deeper planning make the grandmaster more "rational" or just more computationally powerful? Is there a philosophical difference?

Write your response here

Part 2: The Problem of Radical Novelty

Goal: Examine the limits of planning under uncertainty.

A young person choosing a career faces a problem: the career that would make their future self happiest may require skills and preferences they have not yet developed. They cannot plan for a future self whose generative model is unknown.

- How does Active Inference handle planning when the agent's own model will change?
- Does this connect to the philosophical concept of "transformative experience" (L.A. Paul)? Paul argues that some choices (having a child, going to war) transform the chooser so fundamentally that rational planning is impossible.
- Is it rational to choose a transformative experience? How would an Active Inference agent handle this?

Write your response here

Part 3: Teleology and the Meaning of Life

Goal: Explore the deepest philosophical implications of planning.

If planning is inference about the future guided by prior preferences (C vector), then the "purpose" of an agent's life is encoded in its C vector. But where does the C vector come from? Evolution? Culture? Personal experience?

- If prior preferences are products of evolution, does this make "purpose" arbitrary?
- If prior preferences can be changed through learning (Module 06), can an agent choose its own purposes? What philosophical tradition supports this view (hint: existentialism — Sartre)?
- Write a 100-word reflection: Does Active Inference provide a satisfying account of what gives life meaning?

Write your response here

Part 4: Anxiety, Procrastination, and Planning Failure

Goal: Apply the planning framework to everyday psychological phenomena.

Active Inference predicts that anxiety arises when the agent evaluates many possible futures with high precision on negative outcomes, and procrastination arises when the EFE landscape is too flat to distinguish between policies.

- Think of a situation where you felt anxious about the future. Can you reconstruct the "EFE landscape" — many negative futures with high precision?
- Think of a situation where you procrastinated. Was it because you couldn't decide which course of action was best (flat landscape)?
- Do these Active Inference accounts feel psychologically accurate? What do they miss?

Write your response here

Part 5: Synthesis and Course Integration

Goal: Integrate all 8 modules into a unified philosophical vision.

This is the final module. Write a 300-word essay that connects all eight concepts (Systems, Agents, Perception, Cognition, Action, Learning, Communication, Planning) into a coherent philosophical account of what it means to be a mind in the world. Use vocabulary and concepts from at least 5 of the 8 modules.

Write your response here

Lab Summary

Part	Skill Developed	Key Philosophical Move
1	Recursive analysis	Mapping planning depth to sophistication of inference
2	Limit reasoning	Confronting the paradox of planning for a self that does not yet exist
3	Existential analysis	Connecting prior preferences to purpose and meaning
4	Applied phenomenology	Using Active Inference to illuminate everyday psychological phenomena
5	Integrative essay	Synthesizing the full 8-module philosophical framework